

On the Cardinality of Optimal Representations in the Binary-Source Information Bottleneck

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Abstract

The information bottleneck (IB) seeks a representation U of a source X that retains as much information as possible about a target Y , subject to a constraint on $I(U; X)$. A classical argument shows that it suffices to consider representations with at most $|\mathcal{X}| + 1$ symbols, and this bound is known to be tight whenever $|\mathcal{X}| \geq 3$. We show that the binary case behaves differently: if X is binary and Y is finite, then for every joint distribution of (X, Y) and every rate constraint, the IB optimum is attained by a binary U . Hence the bound $|\mathcal{U}| \leq |\mathcal{X}| + 1$ sharpens to $|\mathcal{U}| \leq |\mathcal{X}|$ for binary sources. The proof combines a separating hyperplane argument with the observation that, for a binary source, the ratio of the second derivatives of the two entropy functions involved is concave.

Keywords: Information bottleneck; cardinality bound; binary source; optimal representation; separating hyperplane

1 Introduction

The information bottleneck (IB) [1, 2] formalizes the extraction of relevant information: given a pair (X, Y) on finite alphabets, one looks for a representation U of X , with $U \leftrightarrow X \leftrightarrow Y$, that is as informative about Y as possible while keeping $I(U; X)$ small. Beyond its information-theoretic interest, the IB has served as a lens for analyzing deep neural networks [3, 4] and as an objective for learning compact yet informative representations. For $R \geq 0$, the optimal tradeoff is described by the IB function

$$F(R) = \max \{I(U; Y) : U \leftrightarrow X \leftrightarrow Y, I(U; X) \leq R\}, \quad (1)$$

which is closely related to the conditional entropy bound of Witsenhausen and Wyner [5].

A basic question is how large the alphabet $\mathcal{U}$ must be. Witsenhausen and Wyner [5] first showed that

$$|\mathcal{U}| \leq |\mathcal{X}| + 1$$

suffices; the same bound follows from the support lemma [6]. Gilad-Bachrach et al. [7] claimed that $|\mathcal{U}| \leq |\mathcal{X}|$ suffices.

Benger, Asodeh and Chen [8] pointed out that this claim conflates the IB Lagrangian, for which $|\mathcal{X}|$ symbols do suffice, with the IB function F itself, whose graph may contain linear segments.

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They further showed that the bound $|\mathcal{X}| + 1$ is tight: for Hamming channels with uniform input and $|\mathcal{X}| \geq 3$, $|\mathcal{X}| + 1$ symbols are necessary for all rates in some interval $(0, R_c)$.

For a binary target Y , Tang and Han [9] proposed a bottleneck model that replaces $I(U; Y)$ by the Chernoff information between $P_{U|Y=0}$ and $P_{U|Y=1}$, and sharpened its cardinality bound from $|\mathcal{X}| + 1$ to $k + 1$, where $k \leq |\mathcal{X}|$ is the number of distinct likelihood ratios $P_{X|Y=1}(x)/P_{X|Y=0}(x)$. The same argument yields this bound for the classical IB with binary Y .

The binary case $|\mathcal{X}| = 2$ is not covered by these results. For a doubly symmetric binary source, Mrs. Gerber's Lemma [10] shows that a binary U is optimal, but this relies on the symmetry of the channel. In this paper we show that, for a binary source, a binary U is always optimal:

- For every joint distribution P_{XY} with $\mathcal{X} = \{0, 1\}$ and $\mathcal{Y}$ finite, and every $R \geq 0$, the maximum in (1) is attained by some U with $|\mathcal{U}| \leq 2$ (Theorem 2.1).
- Consequently, the bound $|\mathcal{X}| + 1$ is never tight for binary sources, and the condition $|\mathcal{X}| \geq 3$ in [8] cannot be dropped.

The proof proceeds as follows. We transform the original problem into minimizing $H(Y|U)$ over the convex hull of a curve. By the separating hyperplane theorem, every atom of an optimal U becomes a minimizer of a function ψ . We prove that ψ either has at most two minimizers or is constant. In the first case, we merge the symbols of U into a binary U that remains optimal; in the second case, we find a binary optimal U by time-sharing. The key step is a curvature computation that is specific to binary sources.

The rest of the paper is organized as follows. Section 2 states the setup and the main result, Section 3 collects two lemmas, Section 4 proves the main result, and Section 5 explains why the argument does not extend to $|\mathcal{X}| \geq 3$ and concludes the paper.

2 Problem Setup and Main Result

Throughout, entropies are in nats and $H(\cdot)$ also denotes the entropy of a probability mass function; we follow the notation of [11].

- Let $\mathcal{X} = \{0, 1\}$, and let $P_X = (1 - p, p)$ with $0 < p < 1$ (otherwise everything is trivial).
- Let $W = P_{Y|X}$, and denote its rows by $w_0 = P_{Y|X=0}$ and $w_1 = P_{Y|X=1}$.
- For $0 \leq q \leq 1$, define

$$f(q) = (1 - q)w_0 + qw_1,$$

with entries $f_y(q) = (1 - q)w_0(y) + qw_1(y)$, $y \in \mathcal{Y}$.

- Let $h(q)$ be the binary entropy of q , and $g(q) = H(f(q))$.
- Let $\gamma(q) = (q, h(q), g(q)) : [0, 1] \rightarrow \mathbb{R}^3$, and let K be its convex hull,

$$K = \text{conv } \gamma([0, 1]) = \left\{ \sum_i \lambda_i \gamma(q_i) : \lambda_i \geq 0, \sum_i \lambda_i = 1, q_i \in [0, 1] \right\}.$$

The maximum in (1) is taken over U with finite alphabet $\mathcal{U}$. By the general cardinality bound [5], it is attained by some U with $|\mathcal{U}| \leq |\mathcal{X}| + 1$. Our main result sharpens this bound in the binary case.

Theorem 2.1. *Let $\mathcal{X} = \{0, 1\}$ and let $\mathcal{Y}$ be finite. Then, for any $R \geq 0$, the maximum in (1) is attained by some U with $|\mathcal{U}| \leq 2$. That is, the general cardinality bound $|\mathcal{U}| \leq |\mathcal{X}| + 1$ sharpens to $|\mathcal{U}| \leq |\mathcal{X}|$ in the binary case.*

3 Preliminaries

Lemma 3.1 (Separating hyperplane theorem [12, Section 2.5]). *Let $S_1, S_2 \subset \mathbb{R}^n$ be nonempty, convex, and disjoint. Then there exist a nonzero $c \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$ such that*

$$c \cdot x \geq \alpha \geq c \cdot y, \quad \forall x \in S_1, y \in S_2.$$

Lemma 3.2 (Identity theorem for polynomials [13, Chapter IV]). *Let $P(x)$ be a polynomial. If $P(x) = 0$ on a nondegenerate interval $I \subset \mathbb{R}$, then $P(x) \equiv 0$ on $\mathbb{R}$.*

4 Proof of Theorem 2.1

4.1 Step 1: Reduction to K

Consider a Markov chain $U \leftrightarrow X \leftrightarrow Y$ with $\mathcal{U}$ finite, and let $q_u = P(X = 1|U = u)$. Then, by Markovity,

$$P_{Y|U=u} = P_{Y|X=0}P_{X=0|U=u} + P_{Y|X=1}P_{X=1|U=u} = w_0(1 - q_u) + w_1q_u = f(q_u).$$

Since $P_{X|U=u} = (1 - q_u, q_u)$ and $\sum_u P_U(u)q_u = p$, we derive

$$\sum_u P_U(u)\gamma(q_u) = (p, H(X|U), H(Y|U)) \in K.$$

Conversely, let $\lambda_u \geq 0$ and $q_u \in [0, 1]$, $u = 1, \dots, |\mathcal{U}|$, satisfy

$$\sum_u \lambda_u = 1, \quad \sum_u \lambda_u q_u = p.$$

Setting

$$P_U(u) = \lambda_u, \quad P_{X|U=u} = (1 - q_u, q_u), \quad P_{Y|X,U} = W$$

gives

$$(p, H(X|U), H(Y|U)) = \sum_u \lambda_u \gamma(q_u).$$

Therefore, the achievable pairs $(H(X|U), H(Y|U))$ are exactly the points (s, t) with $(p, s, t) \in K$. Moreover, since

$$I(U; X) = h(p) - H(X|U), \quad I(U; Y) = H(Y) - H(Y|U),$$

and both $h(p)$ and $H(Y)$ are fixed, computing (1) is equivalent to

$$\min \{t : (p, s, t) \in K, s \geq h(p) - R\}.$$

Let U^* be an optimal U with $|\mathcal{U}| \leq |\mathcal{X}| + 1$, which exists by the general cardinality bound. Without loss of generality, $P_{U^*}(u) > 0$ for all u . Let

$$z^* \triangleq (p, s^*, t^*) = (p, H(X|U^*), H(Y|U^*)) \in K$$

be its point, and let $q_u^* = P(X = 1|U^* = u)$.

4.2 Step 2: Separating Hyperplane

Define

$$S_{t^*} \triangleq \{(p, s, t) : s \geq h(p) - R, t < t^*\} \subset \mathbb{R}^3.$$

Then S_{t^*} is convex, since it is a product of intervals.

We first prove $K \cap S_{t^*} = \emptyset$. Suppose some $(p, s, t) \in K \cap S_{t^*}$; then $s \geq h(p) - R$ and $t < t^*$. By Step 1, there is a U such that $H(X|U) = s$ and $H(Y|U) = t$, which implies

$$I(U; X) = h(p) - s \leq R, \quad H(Y|U) = t < t^*, \quad \text{i.e., } I(U; Y) > I(U^*; Y).$$

This contradicts the optimality of U^* . Hence $K \cap S_{t^*} = \emptyset$.

We then prove $z^* \in \overline{S_{t^*}}$. Define

$$z_n^* \triangleq (p, s^*, t^* - \frac{1}{n}), \quad n \geq 1.$$

Because U^* is feasible, we have $s^* \geq h(p) - R$; and because $t^* - \frac{1}{n} < t^*$, we derive $z_n^* \in S_{t^*}$. Since $z_n^* \rightarrow z^*$, we have $z^* \in \overline{S_{t^*}}$.

Applying the separating hyperplane theorem (Lemma 3.1) with $S_1 = K$ and $S_2 = S_{t^*}$, there exist $(c_0, c_1, c_2) \neq 0$ and $\alpha \in \mathbb{R}$ such that

$$c_0 q + c_1 s + c_2 t \geq \alpha \geq c_0 p + c_1 s' + c_2 t', \quad \forall (q, s, t) \in K, (p, s', t') \in S_{t^*}. \quad (2)$$

Since $z^* \in K$, the left inequality gives

$$c_0 p + c_1 s^* + c_2 t^* \geq \alpha;$$

since $z^* \in \overline{S_{t^*}}$, the right inequality gives

$$c_0 p + c_1 s^* + c_2 t^* \leq \alpha.$$

Hence

$$\alpha = c_0 p + c_1 s^* + c_2 t^*.$$

We prove $c_2 \geq 0$. Setting $s' = s^*$ and $t' = t^* - n$ in (2) gives

$$\alpha \geq c_0 p + c_1 s^* + c_2 t^* - c_2 n = \alpha - c_2 n,$$

i.e., $c_2 n \geq 0$ for all $n \geq 1$. Hence $c_2 \geq 0$.

Furthermore, we prove that c_1 and c_2 cannot both be 0. Suppose $c_1 = c_2 = 0$. Then the left inequality of (2) gives

$$c_0 q \geq \alpha = c_0 p, \quad \text{i.e., } c_0(q - p) \geq 0, \quad \forall (q, s, t) \in K.$$

Taking $(q, s, t) = \gamma(0)$ and $\gamma(1)$ gives

$$-c_0 p \geq 0, \quad c_0(1 - p) \geq 0.$$

Because $0 < p < 1$, we have $c_0 = 0$, which contradicts $(c_0, c_1, c_2) \neq 0$.

Let

$$\psi(q) \triangleq c_0 q + c_1 h(q) + c_2 g(q).$$

Applying the left inequality of (2) to $\gamma(q) \in K$ gives

$$\psi(q) \geq \alpha, \quad \forall q \in [0, 1].$$

Averaging over U^* gives

$$\begin{aligned}\sum_u P_{U^*}(u)(\psi(q_u^*) - \alpha) &= c_0 \sum_u P_{U^*}(u)q_u^* + c_1 \sum_u P_{U^*}(u)h(q_u^*) + c_2 \sum_u P_{U^*}(u)g(q_u^*) - \alpha \\ &= c_0 p + c_1 s^* + c_2 t^* - \alpha \\ &= 0.\end{aligned}$$

Since every term on the left is nonnegative, $\psi(q_u^*) = \alpha$ for every u . In a word, every q_u^* is a minimizer of ψ on $[0, 1]$, with minimum value α .

4.3 Step 3: Curvature of ψ

The aim of this step is to show that the set where $\psi'' < 0$ is an interval, and Step 4 uses this to bound the number of minimizers of ψ .

Throughout this step, $q \in (0, 1)$. Without loss of generality, $P_Y(y) > 0$ for all $y \in \mathcal{Y}$, which gives

$$f_y(q) = (1 - q)w_0(y) + qw_1(y) > 0, \quad \forall y \in \mathcal{Y}, \quad q \in (0, 1).$$

Let $d \triangleq w_1 - w_0$, so that $f_y(q) = w_0(y) + qd(y)$. A direct calculation gives

$$h''(q) = -\frac{1}{q(1-q)}, \quad g''(q) = -\sum_y \frac{d(y)^2}{f_y(q)}.$$

Since $(c_0 q)'' = 0$, it follows that

$$\psi''(q) = c_1 h''(q) + c_2 g''(q) = h''(q)(c_1 + c_2 r(q)),$$

where

$$r(q) \triangleq \frac{g''(q)}{h''(q)} = q(1-q) \sum_y \frac{d(y)^2}{f_y(q)}.$$

We now show that r is concave. For each $y \in \mathcal{Y}$, let $\tilde{f}_y(q) \triangleq (1-q)w_1(y) + qw_0(y) = w_1(y) - qd(y)$. Then

$$f_y(q)\tilde{f}_y(q) = (w_0(y) + qd(y))(w_1(y) - qd(y)) = w_0(y)w_1(y) + d(y)^2 q(1-q).$$

Dividing by $f_y(q)$, summing over y , and using $\sum_y \tilde{f}_y(q) = 1$, we obtain

$$r(q) = 1 - \sum_y \frac{w_0(y)w_1(y)}{f_y(q)}. \tag{3}$$

Each $1/f_y(q)$ is convex in q , being the reciprocal of a positive affine function, and $w_0(y)w_1(y) \geq 0$. Hence r is concave on $(0, 1)$.

Finally, since $h''(q) < 0$,

$$\mathcal{Q} \triangleq \{q \in (0, 1) : \psi''(q) < 0\} = \{q \in (0, 1) : c_1 + c_2 r(q) > 0\}.$$

Since $c_2 \geq 0$ (Step 2), the mapping $q \mapsto c_1 + c_2 r(q)$ is concave on $(0, 1)$. Hence $\mathcal{Q}$, a superlevel set of a concave function, is an interval, possibly empty.

4.4 Step 4: At Most Two Minimizers

With ψ and α defined in Step 2, we have the following lemma.

Lemma 4.1. *If $\psi(q) = \alpha$ holds for more than two $q \in [0, 1]$, then $\psi \equiv \alpha$ on $[0, 1]$.*

Proof. Recall from Step 2 that $\psi \geq \alpha$ on $[0, 1]$. Moreover, ψ is continuous on $[0, 1]$ and twice differentiable on $(0, 1)$. Suppose there exist $0 \leq q_1 < q_2 < q_3 \leq 1$ such that

$$\psi(q_1) = \psi(q_2) = \psi(q_3) = \alpha.$$

With $\mathcal{Q}$ defined in Step 3, we first prove that $\mathcal{Q}$ cannot meet both intervals (q_1, q_2) and (q_2, q_3) . If it does, then, since $\mathcal{Q}$ is an interval, we have $q_2 \in \mathcal{Q}$, hence $\psi''(q_2) < 0$. However, $q_2 \in (0, 1)$ and $\psi(q_2) = \alpha = \min \psi$, so q_2 is an interior minimizer of ψ , which forces $\psi''(q_2) \geq 0$, a contradiction.

Therefore, $\mathcal{Q}$ misses one of (q_1, q_2) and (q_2, q_3) . Assume it misses (q_1, q_2) ; the other case is identical. Then $\psi''(q) \geq 0$ on (q_1, q_2) , so ψ is convex on $[q_1, q_2]$. Since $\psi(q_1) = \psi(q_2) = \alpha$, convexity and continuity give $\psi(q) \leq \alpha$ on $[q_1, q_2]$. Combining with $\psi \geq \alpha$, we have

$$\psi(q) = \alpha, \quad \forall q \in [q_1, q_2].$$

Hence $\psi''(q) = 0$ on (q_1, q_2) . Since $\psi''(q) = h''(q)(c_1 + c_2 r(q))$ with $h''(q) \neq 0$, we derive

$$c_1 + c_2 r(q) = 0, \quad \forall q \in (q_1, q_2).$$

Define

$$\mathcal{G}(q) \triangleq (c_1 + c_2 r(q)) \prod_y f_y(q).$$

Substituting (3) gives

$$\begin{aligned} \mathcal{G}(q) &= \left(c_1 + c_2 - c_2 \sum_y \frac{w_0(y)w_1(y)}{f_y(q)} \right) \prod_{y'} f_{y'}(q) \\ &= (c_1 + c_2) \prod_{y'} f_{y'}(q) - c_2 \sum_y w_0(y)w_1(y) \frac{\prod_{y'} f_{y'}(q)}{f_y(q)} \\ &= (c_1 + c_2) \prod_{y'} f_{y'}(q) - c_2 \sum_y w_0(y)w_1(y) \prod_{y' \neq y} f_{y'}(q). \end{aligned}$$

Since each $f_y(q)$ is affine in q , the function $\mathcal{G}(q)$ is a polynomial.

Since $\mathcal{G}(q) = 0$ on the nondegenerate interval (q_1, q_2) , the identity theorem for polynomials (Lemma 3.2) gives $\mathcal{G}(q) \equiv 0$. Because $\prod_y f_y(q) > 0$ on $(0, 1)$, we have $c_1 + c_2 r(q) \equiv 0$ on $(0, 1)$, which gives

$$\psi''(q) = h''(q)(c_1 + c_2 r(q)) = 0, \quad \forall q \in (0, 1).$$

Hence, for some $\zeta_0, \zeta_1 \in \mathbb{R}$,

$$\psi(q) = \zeta_1 q + \zeta_0 \quad \text{on } (0, 1),$$

and by continuity also on $[0, 1]$. Since $\psi(q_1) = \psi(q_2) = \alpha$ and $q_1 < q_2$, we derive $\zeta_1 = 0$ and $\zeta_0 = \alpha$, i.e., $\psi \equiv \alpha$ on $[0, 1]$. $\square$

4.5 Step 5: Construction of a Binary Optimal U

By Step 2, every q_u^* is a minimizer of ψ , i.e., $\psi(q_u^*) = \alpha$. Hence, by Lemma 4.1, one of two cases holds: either the q_u^* take at most two distinct values, or $\psi \equiv \alpha$.

Case 1: the q_u^* take at most two distinct values. Let the distinct values be $\hat{q}_1, \dots, \hat{q}_k$, with $k \in \{1, 2\}$. Merging the symbols of U^* that share the same q_u^* gives a new $\hat{U} = \sigma(U^*)$, where

$$\sigma(u) = j, \quad \text{if } q_u^* = \hat{q}_j, \quad j = 1, \dots, k.$$

We now prove that $\hat{U}$ has the same point z^* . Since $\hat{U}$ is a function of U^* , we still have the Markov chain $\hat{U} \leftrightarrow X \leftrightarrow Y$. For $j = 1, \dots, k$,

$$P_{\hat{U}}(j) = \sum_{u: q_u^* = \hat{q}_j} P_{U^*}(u), \quad P(X = 1 | \hat{U} = j) = \frac{1}{P_{\hat{U}}(j)} \sum_{u: q_u^* = \hat{q}_j} P_{U^*}(u) q_u^* = \hat{q}_j.$$

Then, by Step 1,

$$\begin{aligned} (p, H(X|\hat{U}), H(Y|\hat{U})) &= \sum_j P_{\hat{U}}(j) \gamma(\hat{q}_j) \\ &= \sum_j \sum_{u: q_u^* = \hat{q}_j} P_{U^*}(u) \gamma(q_u^*) \\ &= \sum_u P_{U^*}(u) \gamma(q_u^*) \\ &= z^*. \end{aligned}$$

Therefore, $\hat{U}$ is optimal, with $|\hat{\mathcal{U}}| \leq 2$.

Case 2: $\psi \equiv \alpha$. Then $\psi''(q) \equiv 0$ on $(0, 1)$. Since $\psi''(q) = h''(q)(c_1 + c_2 r(q))$ with $h''(q) \neq 0$, we get $c_1 + c_2 r(q) \equiv 0$. In Step 2 we proved that $c_2 \geq 0$ and that c_1, c_2 cannot both be 0. If $c_2 = 0$, then $c_1 + c_2 r(q) \equiv 0$ forces $c_1 = 0$, a contradiction. Thus $c_2 > 0$.

For any $U \leftrightarrow X \leftrightarrow Y$, Step 1 gives

$$c_0 p + c_1 H(X|U) + c_2 H(Y|U) = \sum_u P_U(u) \psi(q_u) = \alpha.$$

Since $c_2 > 0$, this gives

$$H(Y|U) = \frac{\alpha - c_0 p - c_1 H(X|U)}{c_2},$$

so $H(Y|U)$ is determined by $H(X|U)$. Hence any U with $H(X|U) = s^*$ has $H(Y|U) = t^*$, i.e.,

$$I(U; X) = I(U^*; X) \leq R, \quad I(U; Y) = I(U^*; Y),$$

so it is optimal.

It remains to find a binary U with $H(X|U) = s^*$, which we do by time-sharing. For $\theta \in [0, 1]$, let $T \sim \text{Bern}(\theta)$ and $X' \sim P_X$ be independent of each other and of (X, Y) , and define

$$U_\theta = \begin{cases} X, & T = 1, \\ X', & T = 0. \end{cases}$$

Since X and X' are binary, U_θ is binary. Since U_θ is a function of X and the pair (T, X') , which is independent of (X, Y) , we have $U_\theta \leftrightarrow X \leftrightarrow Y$. Moreover, conditioning on T gives

$$\begin{aligned} P(X = x, U_\theta = u) &= \theta P(X = x, X = u) + (1 - \theta) P(X = x, X' = u) \\ &= \theta P_X(x) \mathbf{1}\{u = x\} + (1 - \theta) P_X(x) P_X(u), \end{aligned}$$

which is affine in θ . Since $H(X|U_\theta) = H(X, U_\theta) - H(U_\theta)$ is a continuous function of this pmf, $H(X|U_\theta)$ is continuous in θ . At the endpoints,

$$H(X|U_1) = H(X|X) = 0, \quad H(X|U_0) = H(X|X') = H(X) = h(p),$$

since $U_1 = X$ and $U_0 = X'$ is independent of X . Since

$$0 \leq s^* = H(X|U^*) \leq H(X) = h(p),$$

the intermediate value theorem gives $\theta^* \in [0, 1]$ with $H(X|U_{\theta^*}) = s^*$. By the above, U_{θ^*} is optimal and binary.

In both cases, the maximum in (1) is attained by some U with $|\mathcal{U}| \leq 2$, which completes the proof of Theorem 2.1. $\blacksquare$

5 Discussion and Conclusion

We showed that, in the information bottleneck with a binary source X and a finite target Y , a binary representation U is always optimal, for every joint distribution and every rate constraint. Thus the general cardinality bound $|\mathcal{U}| \leq |\mathcal{X}| + 1$ sharpens to $|\mathcal{U}| \leq |\mathcal{X}|$ in the binary case. We now explain why the argument does not extend to $|\mathcal{X}| \geq 3$.

For $|\mathcal{X}| \geq 3$, Steps 1 and 2 still go through. Each symbol u corresponds to a point $\mathbf{q}_u = P_{X|U=u}$ of the probability simplex on $\mathcal{X}$, which has dimension $|\mathcal{X}| - 1 \geq 2$, and every $\mathbf{q}_u$ of an optimal U minimizes, on this simplex, a function of the form

$$\psi(\mathbf{q}) = \ell(\mathbf{q}) + c_1 H(\mathbf{q}) + c_2 H(\mathbf{q}W),$$

with ℓ affine and $c_2 \geq 0$. Steps 3 and 4, however, rely on q being a scalar:

- *Curvature.* In the binary case, $\psi'' = h''(c_1 + c_2 r)$ with r concave, so the set where $\psi'' < 0$ is an interval. For $|\mathcal{X}| \geq 3$, the second derivative becomes the Hessian

$$c_1 \nabla_{\mathbf{q}}^2 H(\mathbf{q}) + c_2 \nabla_{\mathbf{q}}^2 H(\mathbf{q}W),$$

and the set where it fails to be positive semidefinite is determined by a generalized eigenvalue problem rather than by the sign of a single concave function.

- *Ordering.* Step 4 uses that, among three minimizers $q_1 < q_2 < q_3$, the middle one lies between the other two. A simplex of dimension at least two has no such ordering, and the target would be at most $|\mathcal{X}|$ minimizers rather than two.

These obstructions are not merely technical. For the Hamming channels in [8], every optimal U needs $|\mathcal{X}| + 1$ symbols at small rates. Together with Theorem 2.1, this shows that the tight cardinality bound, i.e., the smallest M such that $|\mathcal{U}| \leq M$ suffices for every P_{XY} , is

$$M = \begin{cases} 2, & |\mathcal{X}| = 2, \\ |\mathcal{X}| + 1, & |\mathcal{X}| \geq 3. \end{cases}$$

Since symbols with the same $\mathbf{q}_u$ can be merged as in Step 5, the corresponding ψ for these Hamming channels has at least $|\mathcal{X}| + 1$ distinct minimizers; hence no argument of this type can yield $|\mathcal{U}| \leq |\mathcal{X}|$ for all sources with $|\mathcal{X}| \geq 3$. Characterizing the sources with $|\mathcal{X}| \geq 3$ for which $|\mathcal{X}|$ symbols suffice remains an open question.

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